

Literature Review

In the current e-commerce landscape, furniture platforms like **IKEA**, **Wayfair**, and **Pottery Barn** set the standard for digital shopping. Key takeaways from researching these platforms include:

- **Component-Based Architecture:** Modern platforms use frameworks like **Angular** to manage complex UI states (like filtering by material, price, or dimensions) without reloading the page.
- **Visual Consistency:** Furniture shopping is highly visual. Successful sites use high-quality imagery and "Lookbook" features to show products in a real-world context.
- **Searchability:** Advanced search algorithms and categorized navigation are crucial to help users find specific items in a large inventory.

Research Gaps & Challenges :

During the research and initial development of **FurniZone**, several gaps were identified that this project seeks to address or improve:

1. **Performance vs. Image Quality:** Most websites struggle with slow loading times due to high-resolution furniture images. FurniZone aims to leverage Angular's "Lazy Loading" and optimized asset management to maintain speed.
2. **Mobile-First Responsive Design:** Many existing local furniture sites are difficult to navigate on smartphones. Our gap analysis showed a need for a seamless mobile experience for users browsing on the go.
3. **Dynamic Filtering:** A common gap in simple furniture sites is the lack of real-time filtering (e.g., seeing only "Office Chairs" under \$200 instantly). We addressed this using **TypeScript** logic for client-side filtering.
4. **State Management:** Transitioning between categories often leads to losing the user's progress. We focused on maintaining a clean state so users can explore without friction.

3. Suggested Improvements :

- **Augmented Reality (AR):** Integrating an AR feature to let users "place" furniture in their room using their phone camera.
- **Payment Gateway:** Integrating a checkout system (Stripe or PayPal).
- **AI Recommendations:** Suggesting items based on the user's browsing history (e.g., suggesting a lamp if they bought a desk).